

Statistical effect tests

Protected-45 analyses use the same assignments in all 18 conditions. The primary response is RMSE for each amino-acid class; exact-assignment squared-error analyses and tests within each residue–nucleus stratum are also retained. The GROMACS-only analysis restricts preparation to the 150-ps frame and 100-frame average. The restricted-386 analysis uses amino-acid-class RMSE in its six tleap conditions.

The independent implementation reproduced 2,493 test results. Maximum absolute differences were 4.15e-12 for F or t and 1.78e-12 for uncorrected p values; all degrees of freedom matched.

Machine-readable record

The complete 1,824-row ANOVA table is `../outputs/factorial_anova_terms.csv`. The complete 669-row planned-comparison table is `../outputs/planned_contrasts.csv`. Independent validation is recorded in `../outputs/independent_validation.csv` and `../outputs/independent_validation.json`.

Protected 45: full 18-condition ANOVA

Term	H ^N	H α	C α	C β	C'	N
Preparation	$F = 31.85$ $q < .001$	$F = 28.17$ $q < .001$	$F = 42.04$ $q < .001$	$F = 38.05$ $q < .001$	$F = 59.00$ $q < .001$	$F = 52.26$ $q < .001$
Input	$F = 0.53$ $q = .526$	$F = 4.78$ $q = .042$	$F = 4.80$ $q = .041$	$F = 0.08$ $q = .820$	$F = 5.29$ $q = .033$	$F = 4.94$ $q = .035$
Weight transfer	$F = 27.89$ $q < .001$	$F = 51.89$ $q < .001$	$F = 31.83$ $q < .001$	$F = 31.44$ $q < .001$	$F = 126.68$ $q < .001$	$F = 89.08$ $q < .001$
Preparation $\times$ input	$F = 1.38$ $q = .301$	$F = 6.53$ $q = .012$	$F = 13.92$ $q < .001$	$F = 10.56$ $q < .001$	$F = 9.17$ $q < .001$	$F = 24.65$ $q < .001$
Preparation $\times$ transfer	$F = 1.42$ $q = .299$	$F = 1.06$ $q = .380$	$F = 16.22$ $q < .001$	$F = 3.28$ $q = .105$	$F = 30.48$ $q < .001$	$F = 2.14$ $q = .194$
Input $\times$ transfer	$F = 29.94$ $q < .001$	$F = 14.54$ $q < .001$	$F = 13.47$ $q < .001$	$F = 1.09$ $q = .368$	$F = 23.74$ $q < .001$	$F = 1.96$ $q = .194$
Preparation $\times$ input $\times$ transfer	$F = 3.66$ $q = .040$	$F = 3.51$ $q = .043$	$F = 11.56$ $q < .001$	$F = 1.75$ $q = .227$	$F = 35.48$ $q < .001$	$F = 0.21$ $q = .838$

Table 1: Repeated-measures ANOVA across amino-acid classes. Multi-degree-of-freedom terms use Greenhouse–Geisser correction; q is BH-adjusted across all terms and nuclei on this analysis surface.

Protected 45: GROMACS-only ANOVA

Term	H ^N	H α	C α	C β	C'	N
Preparation	$F = 120.62$	$F = 144.53$	$F = 38.24$	$F = 78.91$	$F = 91.38$	$F = 79.41$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Input	$F = 1.89$	$F = 2.53$	$F = 2.74$	$F = 0.35$	$F = 3.27$	$F = 9.52$
	$q = .211$	$q = .159$	$q = .146$	$q = .633$	$q = .107$	$q = .006$
Weight transfer	$F = 26.62$	$F = 45.98$	$F = 42.03$	$F = 20.23$	$F = 123.71$	$F = 56.05$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Preparation $\times$ input	$F = 5.97$	$F = 3.27$	$F = 4.36$	$F = 0.27$	$F = 4.85$	$F = 1.88$
	$q = .016$	$q = .107$	$q = .046$	$q = .738$	$q = .056$	$q = .211$
Preparation $\times$ transfer	$F = 0.44$	$F = 2.03$	$F = 6.60$	$F = 24.34$	$F = 3.08$	$F = 4.53$
	$q = .583$	$q = .211$	$q = .038$	$q < .001$	$q = .143$	$q = .083$
Input $\times$ transfer	$F = 17.20$	$F = 15.12$	$F = 23.30$	$F = 0.07$	$F = 34.00$	$F = 1.85$
	$q < .001$	$q < .001$	$q < .001$	$q = .896$	$q < .001$	$q = .211$
Preparation $\times$ input $\times$ transfer	$F = 3.20$	$F = 2.46$	$F = 2.67$	$F = 12.37$	$F = 0.13$	$F = 0.23$
	$q = .107$	$q = .146$	$q = .146$	$q = .002$	$q = .833$	$q = .738$

Table 2: Repeated-measures ANOVA across amino-acid classes in the 12 GROMACS conditions. Preparation is 1 Frame or MD Average. Multi-degree-of-freedom terms use Greenhouse–Geisser correction; q is BH-adjusted across all terms and nuclei on this surface.

Protected 45: planned comparisons

Comparison	H ^N	H α	C α	C β	C'	N
Weight transfer – no transfer	$\Delta = -.037$	$\Delta = -.038$	$\Delta = -.049$	$\Delta = -.054$	$\Delta = -.140$	$\Delta = -.300$
	lower: 19/19	lower: 19/20	lower: 17/20	lower: 17/19	lower: 20/20	lower: 19/19
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
MD Average – 1 Frame	$\Delta = -.039$	$\Delta = -.025$	$\Delta = -.058$	$\Delta = -.083$	$\Delta = -.066$	$\Delta = -.129$
	lower: 19/19	lower: 20/20	lower: 18/20	lower: 19/19	lower: 20/20	lower: 19/19
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
MOPAC PM7 – Structure	$\Delta = -.007$	$\Delta = -.014$	$\Delta = -.012$	$\Delta = +.014$	$\Delta = -.007$	$\Delta = +.065$
	lower: 15/19	lower: 17/20	lower: 12/20	lower: 16/19	lower: 15/20	lower: 2/19
	$q = .497$	$q = .018$	$q = .252$	$q = .813$	$q = .880$	$q = .177$
MOPAC PM7 – AIMNet2	$\Delta = +.002$	$\Delta = -.011$	$\Delta = -.032$	$\Delta = +.005$	$\Delta = -.067$	$\Delta = -.043$
	lower: 15/19	lower: 19/20	lower: 16/20	lower: 17/19	lower: 17/20	lower: 13/19
	$q = .905$	$q = .123$	$q = .048$	$q = .923$	$q = .018$	$q = .123$
AIMNet2 – Structure	$\Delta = -.009$	$\Delta = -.003$	$\Delta = +.020$	$\Delta = +.009$	$\Delta = +.061$	$\Delta = +.108$
	lower: 12/19	lower: 12/20	lower: 7/20	lower: 8/19	lower: 2/20	lower: 3/19
	$q = .142$	$q = .415$	$q = .036$	$q = .497$	$q = .001$	$q = .018$

Table 3: Planned paired comparisons across amino-acid classes. Δ is comparison-minus-reference mean class RMSE in ppm; counts give classes with lower RMSE for the comparison. q is BH-adjusted across the five comparisons and six nuclei.

Restricted 386: tleap ANOVA

Term	H ^N	H α	C α	C β	C'	N
Input	$F = 14.29$	$F = 10.65$	$F = 40.61$	$F = 29.55$	$F = 66.97$	$F = 38.55$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Weight transfer	$F = 392.61$	$F = 92.14$	$F = 33.84$	$F = 45.09$	$F = 92.27$	$F = 37.62$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Input $\times$ transfer	$F = 76.14$	$F = 73.55$	$F = 30.31$	$F = 18.00$	$F = 60.62$	$F = 9.04$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q = .001$

Table 4: Repeated-measures ANOVA across amino-acid classes in the six tleap conditions. Multi-degree-of-freedom terms use Greenhouse–Geisser correction; q is BH-adjusted across all terms and nuclei on this surface.

Restricted 386: planned comparisons

Comparison	H ^N	H α	C α	C β	C'	N
Weight transfer – no transfer	$\Delta = -.040$	$\Delta = -.030$	$\Delta = -.049$	$\Delta = -.051$	$\Delta = -.072$	$\Delta = -.227$
	lower: 19/19	lower: 19/20	lower: 19/20	lower: 19/19	lower: 20/20	lower: 19/20
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
MOPAC PM7 – Structure	$\Delta = -.012$	$\Delta = -.016$	$\Delta = +.006$	$\Delta = -.010$	$\Delta = -.001$	$\Delta = +.059$
	lower: 17/19	lower: 19/20	lower: 10/20	lower: 11/19	lower: 9/20	lower: 7/20
	$q < .001$	$q < .001$	$q = .290$	$q = .074$	$q = .892$	$q = .073$
MOPAC PM7 – AIMNet2	$\Delta = -.013$	$\Delta = -.010$	$\Delta = -.059$	$\Delta = -.071$	$\Delta = -.071$	$\Delta = -.254$
	lower: 19/19	lower: 18/20	lower: 20/20	lower: 18/19	lower: 20/20	lower: 18/20
	$q < .001$	$q = .025$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
AIMNet2 – Structure	$\Delta = +.001$	$\Delta = -.006$	$\Delta = +.065$	$\Delta = +.061$	$\Delta = +.071$	$\Delta = +.313$
	lower: 10/19	lower: 16/20	lower: 0/20	lower: 1/19	lower: 0/20	lower: 1/20
	$q = .781$	$q = .058$	$q < .001$	$q < .001$	$q < .001$	$q < .001$

Table 5: Planned paired comparisons across amino-acid classes. Δ is comparison-minus-reference mean class RMSE in ppm; counts give classes with lower RMSE for the comparison. q is BH-adjusted across the four comparisons and six nuclei.

Protected 45: exact-assignment ANOVA

Term	H ^N	H α	C α	C β	C'	N
Preparation	$F = 52.83$	$F = 79.12$	$F = 131.45$	$F = 83.22$	$F = 81.63$	$F = 98.05$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Input	$F = 6.99$	$F = 51.02$	$F = 11.02$	$F = 1.28$	$F = 62.22$	$F = 20.96$
	$q = .001$	$q < .001$	$q < .001$	$q = .277$	$q < .001$	$q < .001$
Weight transfer	$F = 64.12$	$F = 188.30$	$F = 72.45$	$F = 60.69$	$F = 279.55$	$F = 190.53$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Preparation $\times$ input	$F = 5.56$	$F = 9.21$	$F = 55.70$	$F = 21.08$	$F = 13.43$	$F = 73.86$
	$q = .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Preparation $\times$ transfer	$F = 0.72$	$F = 3.51$	$F = 51.54$	$F = 6.22$	$F = 58.38$	$F = 1.42$
	$q = .459$	$q = .045$	$q < .001$	$q = .007$	$q < .001$	$q = .254$
Input $\times$ transfer	$F = 53.37$	$F = 68.29$	$F = 49.90$	$F = 5.00$	$F = 44.73$	$F = 8.01$
	$q < .001$	$q < .001$	$q < .001$	$q = .010$	$q < .001$	$q < .001$
Preparation $\times$ input $\times$ transfer	$F = 8.56$	$F = 6.57$	$F = 41.35$	$F = 4.82$	$F = 81.48$	$F = 3.57$
	$q < .001$	$q < .001$	$q < .001$	$q = .005$	$q < .001$	$q = .015$

Table 6: Repeated-measures ANOVA of exact-assignment squared errors. Multi-degree-of-freedom terms use Greenhouse–Geisser correction; q is BH-adjusted across all terms and nuclei on this surface.

Protected 45: exact-assignment GROMACS-only ANOVA

Term	H ^N	H α	C α	C β	C'	N
Preparation	$F = 193.39$	$F = 147.59$	$F = 71.72$	$F = 104.84$	$F = 98.99$	$F = 50.31$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Input	$F = 4.95$	$F = 32.94$	$F = 7.65$	$F = 1.24$	$F = 35.59$	$F = 34.39$
	$q = .012$	$q < .001$	$q = .001$	$q = .303$	$q < .001$	$q < .001$
Weight transfer	$F = 47.46$	$F = 154.03$	$F = 102.66$	$F = 46.54$	$F = 290.05$	$F = 167.23$
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
Preparation $\times$ input	$F = 12.35$	$F = 3.89$	$F = 10.38$	$F = 1.61$	$F = 11.31$	$F = 7.72$
	$q < .001$	$q = .028$	$q < .001$	$q = .229$	$q < .001$	$q = .001$
Preparation $\times$ transfer	$F = 0.69$	$F = 0.00$	$F = 7.90$	$F = 23.40$	$F = 0.39$	$F = 4.25$
	$q = .428$	$q = .947$	$q = .007$	$q < .001$	$q = .545$	$q = .049$
Input $\times$ transfer	$F = 31.06$	$F = 58.13$	$F = 73.84$	$F = 1.51$	$F = 77.90$	$F = 10.48$
	$q < .001$	$q < .001$	$q < .001$	$q = .246$	$q < .001$	$q < .001$
Preparation $\times$ input $\times$ transfer	$F = 6.02$	$F = 1.65$	$F = 3.88$	$F = 8.07$	$F = 2.54$	$F = 4.27$
	$q = .005$	$q = .225$	$q = .028$	$q < .001$	$q = .099$	$q = .024$

Table 7: Repeated-measures ANOVA of exact-assignment squared errors in the 12 GROMACS conditions. Preparation is 1 Frame or MD Average; q is BH-adjusted across all terms and nuclei on this surface.

Protected 45: exact-assignment planned comparisons

Comparison	H ^N	H α	C α	C β	C'	N
Weight transfer – no transfer	$\Delta = -.029$	$\Delta = -.037$	$\Delta = -.049$	$\Delta = -.056$	$\Delta = -.131$	$\Delta = -.290$
	lower: 1702/3038	lower: 1865/3025	lower: 1613/2962	lower: 1491/2598	lower: 1453/2207	lower: 1739/2738
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
MD Average – 1 Frame	$\Delta = -.039$	$\Delta = -.025$	$\Delta = -.057$	$\Delta = -.079$	$\Delta = -.058$	$\Delta = -.129$
	lower: 1998/3038	lower: 1963/3025	lower: 1853/2962	lower: 1644/2598	lower: 1420/2207	lower: 1739/2738
	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$	$q < .001$
MOPAC PM7 – Structure	$\Delta = -.013$	$\Delta = -.020$	$\Delta = -.012$	$\Delta = -.001$	$\Delta = -.028$	$\Delta = +.091$
	lower: 1600/3038	lower: 1765/3025	lower: 1539/2962	lower: 1468/2598	lower: 1259/2207	lower: 1257/2738
	$q < .001$	$q < .001$	$q = .051$	$q = .949$	$q < .001$	$q < .001$
MOPAC PM7 – AIMNet2	$\Delta = -.007$	$\Delta = -.016$	$\Delta = -.028$	$\Delta = -.018$	$\Delta = -.084$	$\Delta = -.018$
	lower: 1593/3038	lower: 1811/3025	lower: 1526/2962	lower: 1443/2598	lower: 1368/2207	lower: 1401/2738
	$q = .057$	$q < .001$	$q < .001$	$q = .214$	$q < .001$	$q = .232$
AIMNet2 – Structure	$\Delta = -.006$	$\Delta = -.004$	$\Delta = +.017$	$\Delta = +.017$	$\Delta = +.057$	$\Delta = +.109$
	lower: 1543/3038	lower: 1464/3025	lower: 1431/2962	lower: 1332/2598	lower: 988/2207	lower: 1176/2738
	$q = .120$	$q = .025$	$q = .002$	$q = .021$	$q < .001$	$q < .001$

Table 8: Planned paired comparisons of exact-assignment squared errors. Δ is the change in pooled RMSE in ppm; counts give assignments with smaller squared error for the comparison. q is BH-adjusted across the five comparisons and six nuclei.

Protected 45: residue–nucleus ANOVA inventory

Term	H ^N	H α	C α	C β	C'	N
Preparation	11/19	10/20	16/20	9/19	13/20	11/19
Input	3/19	10/20	4/20	3/19	8/20	3/19
Weight transfer	9/19	16/20	8/20	9/19	18/20	13/19
Preparation $\times$ input	1/19	1/20	12/20	8/19	3/20	11/19
Preparation $\times$ transfer	0/19	0/20	7/20	4/19	6/20	2/19
Input $\times$ transfer	10/19	11/20	9/20	2/19	8/20	3/19
Preparation $\times$ input $\times$ transfer	4/19	4/20	10/20	6/19	11/20	4/19

Table 9: Number of residue-specific assignment tests with BH-adjusted $q < 0.05$ for each ANOVA term. Adjustment was performed for each named term across all available residue–nucleus strata.

Protected 45: GROMACS-only residue–nucleus ANOVA inventory

Term	H ^N	H α	C α	C β	C'	N
Preparation	15/19	14/20	10/20	12/19	9/20	7/19
Input	2/19	7/20	3/20	2/19	7/20	5/19
Weight transfer	4/19	15/20	10/20	6/19	18/20	11/19
Preparation $\times$ input	0/19	0/20	0/20	0/19	0/20	0/19
Preparation $\times$ transfer	0/19	0/20	0/20	0/19	0/20	0/19
Input $\times$ transfer	5/19	8/20	10/20	4/19	11/20	3/19
Preparation $\times$ input $\times$ transfer	0/19	0/20	0/20	0/19	0/20	0/19

Table 10: Number of residue-specific assignment tests with BH-adjusted $q < 0.05$ for each ANOVA term. Adjustment was performed for each named term across all available residue–nucleus strata.

Protected 45: residue–nucleus planned-comparison inventory

Comparison	H ^N	H α	C α	C β	C'	N
Weight transfer – no transfer	19/19 lower	19/20 lower	17/20 lower	17/19 lower	20/20 lower	19/19 lower
	$q < .05$: 9 lower	$q < .05$: 16 lower	$q < .05$: 8 lower	$q < .05$: 8 lower	$q < .05$: 18 lower	$q < .05$: 13 lower
	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 1 higher	$q < .05$: 0 higher	$q < .05$: 0 higher
MD Average – 1 Frame	19/19 lower	20/20 lower	18/20 lower	19/19 lower	20/20 lower	19/19 lower
	$q < .05$: 15 lower	$q < .05$: 14 lower	$q < .05$: 10 lower	$q < .05$: 12 lower	$q < .05$: 9 lower	$q < .05$: 7 lower
	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher
MOPAC PM7 – Structure	16/19 lower	17/20 lower	13/20 lower	15/19 lower	15/20 lower	2/19 lower
	$q < .05$: 2 lower	$q < .05$: 8 lower	$q < .05$: 2 lower	$q < .05$: 0 lower	$q < .05$: 2 lower	$q < .05$: 0 lower
	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 2 higher
MOPAC PM7 – AIMNet2	15/19 lower	18/20 lower	16/20 lower	17/19 lower	17/20 lower	12/19 lower
	$q < .05$: 2 lower	$q < .05$: 6 lower	$q < .05$: 3 lower	$q < .05$: 4 lower	$q < .05$: 9 lower	$q < .05$: 1 lower
	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 0 higher
AIMNet2 – Structure	13/19 lower	13/20 lower	8/20 lower	8/19 lower	2/20 lower	3/19 lower
	$q < .05$: 1 lower	$q < .05$: 0 lower	$q < .05$: 0 lower	$q < .05$: 0 lower	$q < .05$: 0 lower	$q < .05$: 0 lower
	$q < .05$: 0 higher	$q < .05$: 0 higher	$q < .05$: 1 higher	$q < .05$: 2 higher	$q < .05$: 3 higher	$q < .05$: 3 higher

Table 11: Residue–nucleus comparison inventory. The first line is the number with lower RMSE for the named comparison; the second line counts adjusted differences in each direction. Adjustment was performed for each named contrast across all available residue–nucleus strata.